

DDB

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1 Density dependent RMF model

The lagrangian of the Relativistic Mean Field (RMF) theory with the density dependent couplings is given as

$$\begin{aligned} \mathcal{L} = & \bar{\Psi} [\gamma^\mu (i\partial_\mu - g_\omega \omega_\mu - g_\rho \vec{\tau} \cdot \vec{\rho}_\mu) - (m - g_\sigma \sigma)] \Psi \\ & + \frac{1}{2} \partial_\mu \sigma \partial^\mu \sigma - \frac{1}{2} m_\sigma^2 \sigma^2 - \frac{1}{4} \Omega_{\mu\nu} \Omega^{\mu\nu} + \frac{1}{2} m_\omega \omega_\mu \omega^\mu - \frac{1}{4} R_{\mu\nu} R^{\mu\nu} + \frac{1}{2} m_\rho \vec{\rho}_\mu \vec{\rho}^\mu \end{aligned} \quad (1)$$

In the above Lagrangian, Ψ , σ , ω_μ and $\vec{\rho}_\mu$ denote the baryon quantum field, scalar σ meson, vector ω meson and iso-vector ρ meson fields respectively. The symbol m denotes the masses of baryon and respected mesons. In the mean field approximation for the uniform and static matter only time like components of the meson fields are survived and space like components vanish. The hamiltonian density of the model, where the meson mean fields are denoted by σ , ω_0 and ρ_{03} and the fermion fields are treated as quantum fields, is

$$\mathcal{H}_B = \Psi^\dagger \left[-i\vec{\alpha} \cdot \vec{\nabla} + \beta (m - g_\sigma \sigma) \right] \Psi \quad (2)$$

$$\mathcal{H}_\sigma = \frac{1}{2} m_\sigma^2 \sigma^2 \quad (3)$$

$$\mathcal{H}_\omega = -\frac{1}{2} m_\omega^2 \omega^\mu \omega_\mu + \Psi^\dagger g_\omega \omega_0 \Psi \quad (4)$$

$$\mathcal{H}_\rho = -\frac{1}{2} m_\rho^2 \vec{\rho}^\mu \vec{\rho}_\mu + \Psi^\dagger I_{3b} g_\rho \rho_{03} \Psi \quad (5)$$

The expectation values $\mathcal{H}_j$ s are

$$\langle \mathcal{H}_B \rangle = \frac{\gamma}{(2\pi)^3} \int d^3k E_k^* [1 - \sin^2 \theta_-^B - \sin^2 \theta_+^B] \quad (6)$$

$$\langle \mathcal{H}_\sigma \rangle = \frac{1}{2} m_\sigma^2 \sigma^2 \quad (7)$$

$$\langle \mathcal{H}_\omega \rangle = -\frac{1}{2} m_\omega^2 \omega^2 + g_\omega \omega_0 \langle \Psi^\dagger \Psi \rangle \quad (8)$$

$$\langle \mathcal{H}_\rho \rangle = -\frac{1}{2} m_\rho^2 \rho^2 + g_\rho I_{3b} \rho_{03} \langle \Psi^\dagger \Psi \rangle \quad (9)$$

where $E_k^* = \sqrt{k^2 + M^2}$ and the dynamic mass $M = m - g_\sigma \sigma$. Where the couplings are density dependents and defined as

$$g_\sigma = g_{\sigma 0} e^{-(x^{a_\sigma} - 1)} \quad (10)$$

$$g_\omega = g_{\omega 0} e^{-(x^{a_\omega} - 1)} \quad (11)$$

$$g_\rho = g_{\rho 0} e^{-a_\rho(x-1)} \quad (12)$$

where $x = \rho_B/\rho_0$ and the total baryon density $\rho_B = \sum_B n_B$. The scalar condensate $n_B^s = \langle \bar{\Psi}_B \Psi_B \rangle$ and vector densities $n_B = \langle \Psi_B^\dagger \Psi_B \rangle$ corresponding to the baryon B are defined as,

$$\langle \bar{\Psi}_B \Psi_B \rangle = \frac{\gamma}{(2\pi)^3} \int d^3k \frac{M}{E_k^*} [1 - \sin^2 \theta_-^B - \sin^2 \theta_+^B] \quad (13)$$

$$\langle \Psi_B^\dagger \Psi_B \rangle = \frac{\gamma}{(2\pi)^3} \int d^3k [\sin^2 \theta_-^B - \sin^2 \theta_+^B] \quad (14)$$

The thermodynamic potential Ω is defined as

$$\Omega = \epsilon - \frac{1}{\beta} s - \sum_B \mu_B n_B \quad (15)$$

with energy density $\epsilon = \langle \mathcal{H} \rangle_\beta$ and the entropy density, s is defined as

$$s = - \sum_B \frac{\gamma}{(2\pi)^3} \int d^3k [\sin^2 \theta_-^B \log \sin^2 \theta_-^B + \cos^2 \theta_-^B \log \cos^2 \theta_-^B + (- \rightarrow +)] \quad (16)$$

Extremising the thermodynamic potential with respect to the θ functions and the meson field give the meson field equations and the condensate function are found i.e.

$$\frac{\partial \Omega}{\partial \sigma} = 0 = \frac{\partial \Omega}{\partial \omega} = \frac{\partial \Omega}{\partial \rho} = \frac{\partial \Omega}{\partial \theta_+^B} = \frac{\partial \Omega}{\partial \theta_-^B} \quad (17)$$

Now, taking first

$$\begin{aligned} \frac{\partial \Omega}{\partial \sigma} &= m_\sigma^2 \sigma + \sum_B \frac{\gamma}{(2\pi)^3} \int d^3k \frac{M}{E_k^*} \frac{\partial M}{\partial \sigma} [1 - \sin^2 \theta_-^B - \sin^2 \theta_+^B] \\ &= m_\sigma^2 \sigma - \sum_B \frac{g_\sigma \gamma}{(2\pi)^3} \int d^3k \frac{M}{E_k^*} [1 - \sin^2 \theta_-^B - \sin^2 \theta_+^B] \\ \therefore m_\sigma^2 \sigma &= \sum_B \frac{g_\sigma \gamma}{(2\pi)^3} \int d^3k \frac{M}{E_k^*} [1 - \sin^2 \theta_-^B - \sin^2 \theta_+^B] \\ \text{or } m_\sigma^2 \sigma &= \sum_B g_\sigma n_B^s \end{aligned} \quad (18)$$

$$\begin{aligned} \frac{\partial \Omega}{\partial \omega} &= -m_\omega^2 \omega + \sum_B g_\omega \langle \Psi_B^\dagger \Psi_B \rangle \\ \therefore m_\omega^2 \omega &= \sum_B g_\omega \langle \Psi_B^\dagger \Psi_B \rangle \\ \text{or } m_\omega^2 \omega &= \sum_B g_\omega n_B \end{aligned} \quad (19)$$

$$\begin{aligned} \frac{\partial \Omega}{\partial \rho} &= -m_\rho^2 \rho + \sum_B g_\rho I_{3B} \langle \Psi_B^\dagger \Psi_B \rangle \\ \therefore m_\rho^2 \rho &= \sum_B g_\rho I_{3B} \langle \Psi_B^\dagger \Psi_B \rangle \\ \text{or } m_\rho^2 \rho &= \sum_B g_\rho I_{3B} n_B \end{aligned} \quad (20)$$

and

$$\begin{aligned}
\frac{\partial \Omega}{\partial \theta_-^A} &= \frac{\partial \epsilon}{\partial \theta_-^A} - \frac{1}{\beta} \frac{\delta s}{\delta \theta_-^A} - \sum_B \mu_B \frac{\partial n_B}{\partial \theta_-^A} \\
&= \frac{\gamma}{(2\pi)^3} \int d^3k E_k^* (-\sin 2\theta_-^A) + \sum_B \frac{\gamma}{(2\pi)^3} \int d^3k \frac{M}{E_k^*} \left(-\sigma \frac{\partial g_\sigma}{\partial \rho_B} \frac{\partial \rho_B}{\partial \theta_-^A} \right) [1 - \sin^2 \theta_-^B - \sin^2 \theta_+^B] \\
&\quad + \sum_B \omega n_B \frac{\partial g_\omega}{\partial \rho_B} \frac{\partial \rho_B}{\partial \theta_-^A} + \sum_B g_\omega \omega \frac{\partial n_B}{\partial \theta_-^A} + \sum_B \rho I_{3B} n_B \frac{\partial g_\rho}{\partial \rho_B} \frac{\partial \rho_B}{\partial \theta_-^A} + \sum_B g_\rho I_{3B} \rho \frac{\partial n_B}{\partial \theta_-^A} \\
&\quad + \frac{1}{\beta} \frac{\gamma}{(2\pi)^3} \int d^3k [\sin 2\theta_-^A \log \sin^2 \theta_-^A + \sin 2\theta_-^A - \sin 2\theta_-^A \log \cos^2 \theta_-^A - \sin 2\theta_-^A] \\
&\quad - \sum_B \mu_B \frac{\partial n_B}{\partial \theta_-^A}
\end{aligned} \tag{21}$$

Where $\rho_B = \sum_B n_B$. Using Eq. (14) we find

$$\begin{aligned}
\frac{\partial \rho_B}{\partial \theta_-^A} &= \sum_B \frac{\gamma}{(2\pi)^3} \int d^3k \sin 2\theta_-^B \frac{\partial \theta_-^B}{\partial \theta_-^A} \\
&= \sum_B \frac{\gamma}{(2\pi)^3} \int d^3k \sin 2\theta_-^B \delta_{AB} \\
&= \frac{\gamma}{(2\pi)^3} \int d^3k \sin 2\theta_-^A
\end{aligned} \tag{22}$$

then,

$$\begin{aligned}
\frac{\partial \Omega}{\partial \theta_-^A} &= -\frac{\gamma}{(2\pi)^3} \int d^3k E_k^* \sin 2\theta_-^A - \sum_B \sigma n_B^s \frac{\partial g_\sigma}{\partial \rho_B} \frac{\partial \rho_B}{\partial \theta_-^A} \\
&\quad + \sum_B \omega n_B \frac{\partial g_\omega}{\partial \rho_B} \frac{\partial \rho_B}{\partial \theta_-^A} + \sum_B g_\omega \omega \frac{\partial n_B}{\partial \theta_-^A} + \sum_B \rho I_{3B} n_B \frac{\partial g_\rho}{\partial \rho_B} \frac{\partial \rho_B}{\partial \theta_-^A} + \sum_B g_\rho I_{3B} \rho \frac{\partial n_B}{\partial \theta_-^A} \\
&\quad + \frac{1}{\beta} \frac{\gamma}{(2\pi)^3} \int d^3k \sin 2\theta_-^A \log \tan^2 \theta_-^A - \sum_B \mu_B \frac{\partial n_B}{\partial \theta_-^A} \\
&= -\frac{\gamma}{(2\pi)^3} \int d^3k E_k^* \sin 2\theta_-^A + \frac{\gamma}{(2\pi)^3} \int d^3k \sum_B \left\{ -\sigma n_B^s \frac{\partial g_\sigma}{\partial \rho_B} + \omega n_B \frac{\partial g_\omega}{\partial \rho_B} + \rho I_{3B} n_B \frac{\partial g_\rho}{\partial \rho_B} \right\} \sin 2\theta_-^A \\
&\quad + \frac{1}{\beta} \frac{\gamma}{(2\pi)^3} \int d^3k \sin 2\theta_-^A \log \tan^2 \theta_-^A + \frac{\gamma}{(2\pi)^3} \int d^3k \{-\mu_A + g_\rho I_{3A} \rho + g_\omega \omega\}
\end{aligned} \tag{23}$$

Then

$$\frac{\partial \Omega}{\partial \theta_-^A} = \frac{\gamma}{(2\pi)^3} \int d^3k \left\{ -E_k^* + \Sigma^r + g_\omega \omega + g_\rho I_{3A} \rho - \mu_A + \frac{1}{\beta} \log \tan^2 \theta_-^A \right\} \sin 2\theta_-^A \tag{24}$$

where

$$\Sigma^r = \sum_B \left\{ -\sigma n_B^s \frac{\partial g_\sigma}{\partial \rho_B} + \omega n_B \frac{\partial g_\omega}{\partial \rho_B} + \rho I_{3B} n_B \frac{\partial g_\rho}{\partial \rho_B} \right\} \tag{25}$$

Let's define

$$\mu_B^* = \mu_B - g_\omega \omega - g_\rho I_{3B} \rho - \Sigma^r \tag{26}$$

Then the Eq. (24) becomes

$$\frac{\partial \Omega}{\partial \theta_-^A} = \frac{\gamma}{(2\pi)^3} \int d^3k \left\{ -E_k^* - \mu_B^* + \frac{1}{\beta} \log \tan^2 \theta_-^B \right\} \sin 2\theta_-^B \tag{27}$$

From the Eq. (17), we have

$$\begin{aligned} -E_k^* - \mu_B^* + \frac{1}{\beta} \log \tan^2 \theta_-^B &= 0 \\ \log \tan^2 \theta_-^B &= \beta (E_k^* + \mu_B^*) \\ \tan^2 \theta_-^B &= e^{\beta(E_k^* + \mu_B^*)} \end{aligned}$$

and so

$$\sin^2 \theta_-^B = \frac{1}{1 + e^{\beta(E_k^* - \mu_B^*)}} \equiv f_- \quad (28)$$

and similarly

$$\sin^2 \theta_+^B = \frac{1}{1 + e^{\beta(E_k^* + \mu_B^*)}} \equiv f_+ \quad (29)$$

which are Fermi-distributions for the particles and anti-particles. To get the Equation of State (EOS) we use the thermodynamic relations

$$p = -\Omega \quad (30)$$

$$\epsilon = \frac{1}{\beta} s + \sum_B n_B \mu_B - p \quad (31)$$

We can find the EOS in the model at zero temperature by subtracting out the vacuum contribution. This yields the following expression for the energy density and pressure

$$\epsilon = \sum_B \frac{\gamma}{(2\pi)^3} \int_0^{k_{FB}} d^3k E_k^* + \frac{1}{2} m_\sigma^2 \sigma^2 + \frac{1}{2} m_\omega^2 \omega^2 + \frac{1}{2} m_\rho^2 \rho^2 \quad (32)$$

$$p = \sum_B n_B \mu_B - \epsilon \quad (33)$$

The vector and scalar densities for a particular baryon are defined as

$$n_B = \frac{\gamma k_{FB}^3}{6\pi^2} \quad (34)$$

$$n_B^s = \frac{\gamma}{(2\pi)^3} \int_0^{k_{FB}} \frac{M}{E_k^*} d^3k \quad (35)$$

where γ is the degeneracy, $E_k^* = \sqrt{k^2 + M^2}$ and k_{Fi} is the fermi-momenta of the i th particle.

2 Equation of state

The equation of state of density dependent couplings with bayesian analysis (DDB) RMF is given as following

$$\epsilon = \sum_{i=p,n,e,\mu} \frac{\gamma_i}{(2\pi)^3} \int_0^{k_{Fi}} E_k^* d^3k + \frac{1}{2} m_\sigma^2 \sigma_0^2 + \frac{1}{2} m_\omega^2 \omega_0^2 + \frac{1}{2} m_\rho^2 \rho_{03}^2 \quad (36)$$

$$p = \sum_{i=p,n,e,\mu} \mu_i n_i - \epsilon \quad (37)$$

3 sound velocities

3.1 Equilibrium sound velocity

$$c_e^2 = \frac{\partial p}{\partial \epsilon} \quad (38)$$

3.2 Adiabatic sound velocity

$$c_s^2 = \frac{\left(\frac{\partial p}{\partial \rho_B}\right)_{y_i}}{\left(\frac{\partial \epsilon}{\partial \rho_B}\right)_{y_i}} \quad (39)$$

$$\begin{aligned} \left(\frac{\partial p}{\partial \rho_B}\right)_{y_i} &= \sum_{i=p,n,e,\mu} \left(\frac{\partial \mu_i}{\partial \rho_B}\right)_{y_i} n_i + \sum_{i=p,n,e,\mu} \mu_i \left(\frac{\partial n_i}{\partial \rho_B}\right)_{y_i} - \left(\frac{\partial \epsilon}{\partial \rho_B}\right)_{y_i} \\ &= \sum_{i=p,n,e,\mu} \left(\frac{\partial \mu_i}{\partial \rho_B}\right)_{y_i} n_i + \sum_{i=p,n,e,\mu} \mu_i y_i - \left(\frac{\partial \epsilon}{\partial \rho_B}\right)_{y_i} \end{aligned} \quad (40)$$

since

$$\mu_i^* = \mu_i - g_\omega \omega - g_\rho I_{3i} \rho - \Sigma^r \quad (41)$$

then

$$\begin{aligned} \left(\frac{\partial \mu_i}{\partial \rho_B}\right)_{y_i} &= \left(\frac{\partial \mu_i^*}{\partial \rho_B}\right)_{y_i} + \left(\frac{\partial g_\omega}{\partial \rho_B}\right)_{y_i} \omega + g_\omega \left(\frac{\partial \omega}{\partial \rho_B}\right)_{y_i} \\ &\quad + \left(\frac{\partial g_\rho}{\partial \rho_B}\right)_{y_i} I_{3i} \rho + g_\rho I_{3i} \left(\frac{\partial \rho}{\partial \rho_B}\right)_{y_i} + \left(\frac{\partial \Sigma^r}{\partial \rho_B}\right)_{y_i} \end{aligned} \quad (42)$$

where $\mu_i^* = \sqrt{k_{Fi}^2 + M^2}$ and so

$$\left(\frac{\partial \mu_i^*}{\partial \rho_B}\right)_{y_i} = \frac{1}{\mu_i^*} \left[k_{Fi} \left(\frac{\partial k_{Fi}}{\partial \rho_B}\right)_{y_i} + M \left(\frac{\partial M}{\partial \rho_B}\right)_{y_i} \right] \quad (43)$$

and the first derivatives of the density dependent couplings are given as

$$\left(\frac{\partial g_\sigma}{\partial \rho_B}\right)_{y_i} = -\frac{g_\sigma a_\sigma}{\rho_0} x^{a_\sigma-1} \quad (44)$$

$$\left(\frac{\partial g_\omega}{\partial \rho_B}\right)_{y_i} = -\frac{g_\omega a_\omega}{\rho_0} x^{a_\omega-1} \quad (45)$$

$$\left(\frac{\partial g_\rho}{\partial \rho_B}\right)_{y_i} = -\frac{g_\rho a_\rho}{\rho_0} \quad (46)$$

and their second derivatives used to calculate $\left(\frac{\partial \Sigma^r}{\partial \rho_B}\right)_{y_i}$

$$\left(\frac{\partial^2 g_\sigma}{\partial \rho_B^2}\right)_{y_i} = -\left(\frac{\partial g_\sigma}{\partial \rho_B}\right)_{y_i} \frac{a_\sigma}{\rho_0} x^{a_\sigma-1} - \frac{g_\sigma a_\sigma (a_\sigma - 1)}{\rho_0^2} x^{a_\sigma-2} \quad (47)$$

$$= -\left(\frac{\partial g_\sigma}{\partial \rho_B}\right)_{y_i} \frac{a_\sigma}{\rho_0} x^{a_\sigma-1} + \left(\frac{\partial g_\sigma}{\partial \rho_B}\right)_{y_i} \frac{(a_\sigma - 1)}{x \rho_0} \quad (48)$$

$$= \left(\frac{\partial g_\sigma}{\partial \rho_B}\right)_{y_i} \frac{-a_\sigma x^{a_\sigma} + a_\sigma - 1}{x \rho_0} \quad (49)$$

$$\left(\frac{\partial^2 g_\omega}{\partial \rho_B^2}\right)_{y_i} = -\left(\frac{\partial g_\omega}{\partial \rho_B}\right)_{y_i} \frac{a_\omega}{\rho_0} x^{a_\omega-1} - \frac{g_\omega a_\omega (a_\omega - 1)}{\rho_0^2} x^{a_\omega-2} \quad (50)$$

$$= \left(\frac{\partial g_\omega}{\partial \rho_B}\right)_{y_i} \frac{-a_\omega x^{a_\omega} + a_\omega - 1}{x \rho_0} \quad (51)$$

$$\left(\frac{\partial^2 g_\rho}{\partial \rho_B^2}\right)_{y_i} = -\left(\frac{\partial g_\rho}{\partial \rho_B}\right)_{y_i} \frac{a_\rho}{\rho_0} \quad (52)$$

Now using Eq. (25), we find

$$\begin{aligned} \left(\frac{\partial \Sigma^r}{\partial \rho_B}\right)_{y_i} = \sum_{i=p,n} & \left[-\sigma n_i^s \left(\frac{\partial^2 g_\sigma}{\partial \rho_B^2}\right)_{y_i} - \sigma \left(\frac{\partial n_i^s}{\partial \rho_B}\right)_{y_i} \left(\frac{\partial g_\sigma}{\partial \rho_B}\right)_{y_i} - \left(\frac{\partial \sigma}{\partial \rho_B}\right)_{y_i} n_i^s \left(\frac{\partial g_\sigma}{\partial \rho_B}\right)_{y_i} \right. \\ & + \omega n_i \left(\frac{\partial^2 g_\omega}{\partial \rho_B^2}\right)_{y_i} + \omega \left(\frac{\partial n_i}{\partial \rho_B}\right)_{y_i} \left(\frac{\partial g_\omega}{\partial \rho_B}\right)_{y_i} + \left(\frac{\partial \omega}{\partial \rho_B}\right)_{y_i} n_i \left(\frac{\partial g_\omega}{\partial \rho_B}\right)_{y_i} \\ & \left. + \rho I_{3i} n_i \left(\frac{\partial^2 g_\rho}{\partial \rho_B^2}\right)_{y_i} + \rho I_{3i} \left(\frac{\partial n_i}{\partial \rho_B}\right)_{y_i} \left(\frac{\partial g_\rho}{\partial \rho_B}\right)_{y_i} + \left(\frac{\partial \rho}{\partial \rho_B}\right)_{y_i} I_{3i} n_i \left(\frac{\partial g_\rho}{\partial \rho_B}\right)_{y_i} \right] \quad (53) \end{aligned}$$

The scalar density of i th baryon is defined as

$$\begin{aligned} n_i^s &= \frac{\gamma}{(2\pi)^3} \int_0^{k_{Fi}} \frac{M}{E_k^*} d^3k \\ &= \frac{\gamma M}{(2\pi)^3} \frac{4\pi}{4} \left\{ 2k_{Fi} E_k^* - M^2 \log \left[\frac{E_k^* + k_{Fi}}{E_k^* - k_{Fi}} \right] \right\} \\ &= \frac{\gamma M}{4\pi^2} \left\{ k_{Fi} E_k^* - M^2 \log \left[\frac{E_k^* + k_{Fi}}{M} \right] \right\} \quad (54) \end{aligned}$$

then

$$\begin{aligned} \left(\frac{\partial n_i^s}{\partial \rho_B}\right)_{y_i} &= \frac{n_i^s}{M} \left(\frac{\partial M}{\partial \rho_B}\right)_{y_i} + \frac{\gamma M}{4\pi^2} \left(\frac{\partial}{\partial n_B} \left\{ k_{Fi} E_k^* - M^2 \log \left[\frac{E_k^* + k_{Fi}}{M} \right] \right\}\right)_{y_i} \\ &= \frac{n_i^s}{M} \left(\frac{\partial M}{\partial \rho_B}\right)_{y_i} + \frac{\gamma M}{4\pi^2} \left\{ \left(\frac{\partial k_{Fi}}{\partial \rho_B}\right)_{y_i} E_k^* + k_{Fi} \left(\frac{\partial E_k^*}{\partial \rho_B}\right)_{y_i} \right. \\ &\quad \left. - 2M \left(\frac{\partial M}{\partial \rho_B}\right)_{y_i} \log z_i - \frac{M^2}{z_i} \left(\frac{\partial z_i}{\partial \rho_B}\right)_{y_i} \right\} \quad (55) \end{aligned}$$

where $z_i = \frac{E_k^* + k_{Fi}}{M}$ and $E_k^* = \sqrt{k_{Fi}^2 + M^2}$. Some more definitions are as follows,

$$\left(\frac{\partial E_k^*}{\partial \rho_B}\right)_{y_i} = \frac{1}{E_k^*} \left\{ k_{Fi} \left(\frac{\partial k_{Fi}}{\partial \rho_B}\right)_{y_i} + M \left(\frac{\partial M}{\partial \rho_B}\right)_{y_i} \right\} \quad (56)$$

$$\left(\frac{\partial M}{\partial \rho_B}\right)_{y_i} = -\left(\frac{\partial g_\sigma}{\partial \rho_B}\right)_{y_i} \sigma - g_\sigma \left(\frac{\partial \sigma}{\partial \rho_B}\right)_{y_i} \quad (57)$$

$$\left(\frac{\partial k_{Fi}}{\partial \rho_B}\right)_{y_i} = \frac{k_{Fi}}{3\rho_B} \quad (58)$$

and

$$\begin{aligned}
\left(\frac{\partial z_i}{\partial \rho_B}\right)_{y_i} &= \frac{1}{M} \left[\left(\frac{\partial E_k^*}{\partial \rho_B}\right)_{y_i} + \left(\frac{\partial k_{Fi}}{\partial \rho_B}\right)_{y_i} \right] - \frac{z_i}{M} \left(\frac{\partial M}{\partial \rho_B}\right)_{y_i} \\
&= \frac{1}{M} \left[\frac{1}{E_k^*} \left\{ k_{Fi} \left(\frac{\partial k_{Fi}}{\partial \rho_B}\right)_{y_i} + M \left(\frac{\partial M}{\partial \rho_B}\right)_{y_i} \right\} + \left(\frac{\partial k_{Fi}}{\partial \rho_B}\right)_{y_i} - z_i \left(\frac{\partial M}{\partial \rho_B}\right)_{y_i} \right] \\
&= \frac{1}{M} \left[\left\{ \frac{k_{Fi}}{E_k^*} + 1 \right\} \left(\frac{\partial k_{Fi}}{\partial \rho_B}\right)_{y_i} + \left\{ \frac{M}{E_k^*} - z_i \right\} \left(\frac{\partial M}{\partial \rho_B}\right)_{y_i} \right] \tag{59}
\end{aligned}$$

substituting these definitions in Eq. (55), then we get

$$\begin{aligned}
\left(\frac{\partial n_i^s}{\partial \rho_B}\right)_{y_i} &= \frac{n_i^s}{M} \left(\frac{\partial M}{\partial \rho_B}\right)_{y_i} + \frac{\gamma M}{4\pi^2} \left\{ \left(\frac{\partial k_{Fi}}{\partial \rho_B}\right)_{y_i} E_k^* + k_{Fi} \left(\frac{\partial E_k^*}{\partial \rho_B}\right)_{y_i} \right. \\
&\quad \left. - 2M \left(\frac{\partial M}{\partial \rho_B}\right)_{y_i} \log z_i - \frac{M^2}{z_i} \left(\frac{\partial z_i}{\partial \rho_B}\right)_{y_i} \right\} \\
&= \frac{n_i^s}{M} \left(\frac{\partial M}{\partial \rho_B}\right)_{y_i} + \frac{\gamma M}{4\pi^2} \left\{ \left(\frac{\partial k_{Fi}}{\partial \rho_B}\right)_{y_i} E_k^* + \frac{k_{Fi}}{E_k^*} \left\{ k_{Fi} \left(\frac{\partial k_{Fi}}{\partial \rho_B}\right)_{y_i} + M \left(\frac{\partial M}{\partial \rho_B}\right)_{y_i} \right\} \right. \\
&\quad \left. - 2M \left(\frac{\partial M}{\partial \rho_B}\right)_{y_i} \log z_i - \frac{M}{z_i} \left[\left\{ \frac{k_{Fi}}{E_k^*} + 1 \right\} \left(\frac{\partial k_{Fi}}{\partial \rho_B}\right)_{y_i} + \left\{ \frac{M}{E_k^*} - z_i \right\} \left(\frac{\partial M}{\partial \rho_B}\right)_{y_i} \right] \right\} \\
&= \frac{\gamma M}{4\pi^2} \left(E_k^* + \frac{k_{Fi}^2}{E_k^*} - \frac{M}{z_i} \left\{ \frac{k_{Fi}}{E_k^*} + 1 \right\} \right) \left(\frac{\partial k_{Fi}}{\partial \rho_B}\right)_{y_i} \\
&\quad + \left[\frac{n_i^s}{M} + \frac{\gamma M}{4\pi^2} \left(\frac{M k_{Fi}}{E_k^*} - 2M \log z_i - \frac{M}{z_i} \left\{ \frac{M}{E_k^*} - z_i \right\} \right) \right] \left(\frac{\partial M}{\partial \rho_B}\right)_{y_i} \\
&= A_i \left(\frac{\partial k_{Fi}}{\partial \rho_B}\right)_{y_i} + B_i \left(\frac{\partial M}{\partial \rho_B}\right)_{y_i} \\
&= \frac{A_i k_{Fi}}{3\rho_B} + B_i \left(- \left(\frac{\partial g_\sigma}{\partial \rho_B}\right)_{y_i} \sigma - g_\sigma \left(\frac{\partial \sigma}{\partial \rho_B}\right)_{y_i} \right) \\
&= C_i + D_i \left(\frac{\partial \sigma}{\partial \rho_B}\right)_{y_i} \tag{60}
\end{aligned}$$

where

$$A_i = \frac{\gamma M}{4\pi^2} \left(E_k^* + \frac{k_{Fi}^2}{E_k^*} - \frac{M}{z_i} \left\{ \frac{k_{Fi}}{E_k^*} + 1 \right\} \right) \tag{61}$$

$$B_i = \left[\frac{n_i^s}{M} + \frac{\gamma M}{4\pi^2} \left(\frac{M k_{Fi}}{E_k^*} - 2M \log z_i - \frac{M}{z_i} \left\{ \frac{M}{E_k^*} - z_i \right\} \right) \right] \tag{62}$$

$$C_i = \frac{A_i k_{Fi}}{3\rho_B} - B_i \sigma \left(\frac{\partial g_\sigma}{\partial \rho_B}\right)_{y_i} \tag{63}$$

$$D_i = -B_i g_\sigma \tag{64}$$

Now derivatives of the meson fields from Eqs. (??, ?? and ??),

$$\begin{aligned}
m_\sigma^2 \left(\frac{\partial \sigma}{\partial \rho_B}\right)_{y_i} &= g_\sigma (C_p + C_n) + \frac{\partial g_\sigma}{\partial \rho_B} (n_n^s + n_p^s) + g_\sigma (D_p + D_n) \left(\frac{\partial \sigma}{\partial \rho_B}\right)_{y_i} \\
\therefore \left(\frac{\partial \sigma}{\partial \rho_B}\right)_{y_i} &= \frac{g_\sigma (C_p + C_n) + \frac{\partial g_\sigma}{\partial \rho_B} (n_n^s + n_p^s)}{m_\sigma^2 - g_\sigma (D_p + D_n)} \tag{65}
\end{aligned}$$

$$m_\omega^2 \left(\frac{\partial \omega}{\partial \rho_B} \right)_{y_i} = g_\omega(y_p + y_n) + \frac{\partial g_\omega}{\partial \rho_B} (n_n + n_p) \quad (66)$$

$$m_\rho^2 \left(\frac{\partial \rho}{\partial \rho_B} \right)_{y_i} = \frac{g_\rho}{2}(y_p - y_n) + \frac{1}{2} \frac{\partial g_\rho}{\partial \rho_B} (n_n - n_p) \quad (67)$$

with there definitions we can find the $\left(\frac{\partial \epsilon}{\partial \rho_B} \right)_{y_i}$.

$$\begin{aligned} \epsilon &= \sum_{i=p,n,e,\mu} \frac{\gamma_i}{(2\pi)^3} \int_0^{k_{Fi}} E_k^* d^3k + \frac{1}{2} m_\sigma^2 \sigma^2 + \frac{1}{2} m_\omega^2 \omega^2 + \frac{1}{2} m_\rho^2 \rho^2 \\ &= \sum_{i=p,n,e,\mu} \frac{\gamma_i}{(2\pi)^3} \frac{4\pi}{8} [2E_k^* k_{Fi}^3 + E_k^{*3} k_{Fi} - M^4 \log z_i] + \frac{1}{2} m_\sigma^2 \sigma^2 + \frac{1}{2} m_\omega^2 \omega^2 + \frac{1}{2} m_\rho^2 \rho^2 \\ \left(\frac{\partial \epsilon}{\partial \rho_B} \right)_{y_i} &= \sum_{i=p,n,e,\mu} \frac{\gamma_i}{(2\pi)^3} \frac{4\pi}{8} \left[(6E_k^* k_{Fi}^2 + E_k^{*3}) \left(\frac{\partial k_{Fi}}{\partial \rho_B} \right) + (2k_{Fi}^3 + 3E_k^{*2} k_{Fi}) \left(\frac{\partial E_k^*}{\partial \rho_B} \right) \right. \\ &\quad \left. - 4M^3 \left(\frac{\partial M}{\partial \rho_B} \right)_{y_i} \log z_i - \frac{M^4}{z_i} \left(\frac{\partial z_i}{\partial \rho_B} \right)_{y_i} \right] \\ &\quad + m_\sigma^2 \sigma \left(\frac{\partial \sigma}{\partial \rho_B} \right)_{y_i} + m_\omega^2 \omega \left(\frac{\partial \omega}{\partial \rho_B} \right)_{y_i} + m_\rho^2 \rho \left(\frac{\partial \rho}{\partial \rho_B} \right)_{y_i} \end{aligned} \quad (68)$$

Substituting Eqs. (40 and 68) into Eq. (39) we can find the adiabatic sound speed.